

# Converting Numbers

Week 4

## Objectives

- Know and demonstrate how to convert number from one system to another:
  - Binary to octal to binary
  - Binary to hexadecimal to binary
  - Octal to Decimal to Octal
  - Octal to Hexadecimal to Octal
  - Hexadecimal to Decimal to Hexadecimal

## Binary, Octal and Hexadecimal to Decimal Number

### Binary → Decimal Conversion

In decimal number system, each digit has weight & location. For example, the weight of each digit of 123 depends on its location with respect to the decimal point of the number.

|             | Before the decimal point |      |        |        |        | After     |
|-------------|--------------------------|------|--------|--------|--------|-----------|
| Position, n | n                        | ...  | 1      | 2      | 3      | 1         |
| Weight      | $10^{n-1}$               | ...  | $10^2$ | $10^1$ | $10^0$ | $10^{-1}$ |
| Value       | $10^{n-1}$               | .... | 100    | 10     | 1      |           |

From the table above, the final value of each digit = (digit) \* (value of the weight), therefore:

$$1^{\text{st}} \text{ digit} = 1 * 10^2 = 1 * 100 = 100$$

$$2^{\text{nd}} \text{ digit} = 2 * 10^1 = 2 * 10 = 20$$

$$3^{\text{rd}} \text{ digit} = 3 * 10^0 = 3 * 1 = 3$$

$$123 = 100 + 20 + 3$$

## Binary → Decimal Conversion

Similarly, each digit of a binary number has weight & location. For example, the weight of each digit in **11001** vary depending on its position and location with respect to the decimal point.

|          | Before the decimal point |       |       |       |       | After    |
|----------|--------------------------|-------|-------|-------|-------|----------|
| Position | 5                        | 4     | 3     | 2     | 1     | 1        |
| Weight   | $2^4$                    | $2^3$ | $2^2$ | $2^1$ | $2^0$ | $2^{-1}$ |
| Value    | 16                       | 8     | 4     | 2     | 1     | 0.5      |

To convert, multiply each digit to its corresponding weight and get their sum. So,

$$\begin{aligned}
 (11001)_2 &= 1 * 2^4 + 1 * 2^3 + 0 * 2^2 + 0 * 2^1 + 1 * 2^0 \\
 &= 1 * 16 + 1 * 8 + 0 * 4 + 0 * 2 + 1 * 1 \\
 &= 16 + 8 + 0 + 0 + 1 \\
 &= (25)_{10}
 \end{aligned}$$

## Binary → Decimal Conversion

|          | Before the decimal point |          |      |       |       | After    |          |          |
|----------|--------------------------|----------|------|-------|-------|----------|----------|----------|
| Position | n                        | 11       | .... | 2     | 1     | 1        | 2        | 3        |
| Weight   | $2^{n-1}$                | $2^{10}$ | .... | $2^1$ | $2^0$ | $2^{-1}$ | $2^{-2}$ | $2^{-3}$ |
| Value    | $2^{n-1}$                | 1024     | .... | 2     | 0     | 0.5      | 0.25     | 0.125    |

**Example 1:** Convert  $(10011010010.001)_2$  to Decimal

**Solution:**

$$\begin{aligned}
 (10011010010.001)_2 &= 1 * 2^{10} + 1 * 2^7 + 1 * 2^6 + 1 * 2^4 + 1 * 2^1 \\
 &\quad + 0/2^1 + 0/2^2 + 1/2^3 \\
 (10011010010.001)_2 &= 1024 + 128 + 64 + 16 + 2 + 0.125 \\
 &= (1234.125)_{10}
 \end{aligned}$$

## Binary → Decimal Conversion

**Example 2:** Convert  $(11011)_2$

**Solution:**

$$\begin{array}{ccccccc} (11011)_2 & & & & & & \\ (1 & 1 & 0 & 1 & 1)_2 & & \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 2^4 & + & 2^3 & + & 2^1 & + & 2^0 \quad \leftarrow \text{Weight} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 16 & + & 8 & + & 2 & + & 1 = (27)_{10} \end{array}$$

## Binary → Decimal Conversion

**Example 3:** Convert  $(1.101)_2$

**Solution:**

$$\begin{array}{ccccccc} (1.101)_2 & & & & & & \\ (1 & . & 1 & 0 & 1)_2 & & \\ \downarrow & & \downarrow & & \downarrow & & \\ 2^0 & + & 2^{-1} & + & 2^{-3} & & \leftarrow \text{Weight} \\ \downarrow & & \downarrow & & \downarrow & & \\ 1 & + & 0.5 & + & 0.125 & = & (1.625)_{10} \end{array}$$

## Binary → Decimal Conversion

**Example 4:** Convert  $(1101.101)_2$  to Decimal

**Solution:**

$$\begin{aligned}(1101)_2 &= 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 \\ &= 1 \cdot 8 + 1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 \\ &= 8 + 4 + 0 + 1 \\ &= (13)_{10}\end{aligned}$$

$$\begin{aligned}(0.101)_2 &= 1/2^1 + 0/2^2 + 1/2^3 \\ &= 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} \\ &= 0.5 + 0 + 0.125 \\ &= (0.625)_{10}\end{aligned}$$

So,  $(1101.101)_2 = (13.625)_{10}$

## Octal → Decimal Conversion

|          | Before the decimal point |      |       |       | After    |
|----------|--------------------------|------|-------|-------|----------|
| Position | N                        | .... | 2     | 1     | 1        |
| Weight   | $8^{n-1}$                | .... | $8^1$ | $8^0$ | $8^{-1}$ |
| Value    | $8^{n-1}$                | .... | 8     | 1     | 0.125    |

To convert from octal to decimal, multiply each digit to its corresponding weight and get their sum.

**Example 1:** Convert  $(2322.1)_8$  to Decimal

**Solution:**

$$\begin{aligned}(2322.1)_8 &= 2 \cdot 8^3 + 3 \cdot 8^2 + 2 \cdot 8^1 + 2 \cdot 8^0 + 1 \cdot 8^{-1} \\ (2322.1)_8 &= 2 \cdot 512 + 3 \cdot 64 + 2 \cdot 8 + 2 \cdot 1 + 0.125 \\ (2322.1)_8 &= (1234.125)_{10}\end{aligned}$$

## Octal → Decimal Conversion

*Example 2:* Convert  $(710)_8$  to Decimal

*Solution:*

$$\begin{aligned}(710)_8 &= 7 \cdot 8^2 + 1 \cdot 8^1 + 0 \cdot 8^0 \\ &= 7 \cdot 64 + 1 \cdot 8 + 0 \\ &= (456)_{10}\end{aligned}$$

## Hexa → Decimal Conversion

|          | Before the decimal point |      |        |        | After     |
|----------|--------------------------|------|--------|--------|-----------|
| Position | n                        | .... | 2      | 1      | 1         |
| Weight   | $16^{n-1}$               | .... | $16^1$ | $16^0$ | $16^{-1}$ |
| Value    | $16^{n-1}$               | .... | 16     | 1      | 0.0625    |

To convert from Hex to decimal, multiply each digit to its corresponding weight and get their sum.

**Example 1:** Convert  $(4D2.2)_{16}$  to Decimal (note: D=13)

**Solution:**

$$\begin{aligned}(4D2.2)_{16} &= 4 \cdot 16^2 + 13 \cdot 16^1 + 2 \cdot 16^0 + 2 \cdot 16^{-1} \\ (4D2.2)_{16} &= 4 \cdot 256 + 13 \cdot 16 + 2 \cdot 1 + 2 \cdot 0.0625 \\ (4D2.2)_{16} &= 1024 + 208 + 2 + 0.125 \\ (4D2.2)_{16} &= (1234.125)_{10}\end{aligned}$$

## Hexa → Decimal Conversion

*Example 2:*  $(4A1)_{16}$  to Decimal

*Solution:*

$$\begin{aligned}(4A1)_{16} &= 4 \cdot 16^2 + 10 \cdot 16^1 + 1 \cdot 16^0 \\ &= 1024 + 160 + 1 \\ &= (1185)_{10}\end{aligned}$$

Binary → Octal → Binary

## Binary → Octal → Binary

| DEC | BIN | OCT |
|-----|-----|-----|
| 0   | 000 | 0   |
| 1   | 001 | 1   |
| 2   | 010 | 2   |
| 3   | 011 | 3   |
| 4   | 100 | 4   |
| 5   | 101 | 5   |
| 6   | 110 | 6   |
| 7   | 111 | 7   |

- The mathematical method:
  - BIN → DEC
  - DEC → OCT
- The shortcut method is to cluster the binary number with 3-bit binary strings from the decimal point & then look up at the table to get its OCT equivalent.

## Binary → Octal → Binary

| DEC | BIN | OCT |
|-----|-----|-----|
| 0   | 000 | 0   |
| 1   | 001 | 1   |
| 2   | 010 | 2   |
| 3   | 011 | 3   |
| 4   | 100 | 4   |
| 5   | 101 | 5   |
| 6   | 110 | 6   |
| 7   | 111 | 7   |

- In the event that there is a bit string with less than 3 digits, the necessary number of 0's is added and the string is considered as a 3-bit string.

## Binary → Octal → Binary

| DEC | BIN | OCT |
|-----|-----|-----|
| 0   | 000 | 0   |
| 1   | 001 | 1   |
| 2   | 010 | 2   |
| 3   | 011 | 3   |
| 4   | 100 | 4   |
| 5   | 101 | 5   |
| 6   | 110 | 6   |
| 7   | 111 | 7   |

Example 1:

Convert  $(237)_8$  to Decimal

Solution (Short-cut):

$$\begin{aligned}(237)_8 &= (010\ 011\ 111)_2 \\ &= (10\ 011\ 111)_2\end{aligned}$$

## Binary → Octal → Binary

Example 1: Convert  $(237)_8$  to Binary

Solution (Mathematical):

1. Convert OCT → DEC

$$(237)_8 = 2 \cdot 8^2 + 3 \cdot 8^1 + 7 \cdot 8^0 = 2 \cdot 64 + 3 \cdot 8 + 7 \cdot 1$$

$$(237)_8 = (159)_{10}$$

2. Convert the resulting decimal value  $(159)_{10}$  to binary

## Binary → Octal → Binary

2 | 159 | Rem

2 | 79 | 1

2 | 39 | 1

2 | 19 | 1

2 | 9 | 1

2 | 4 | 1

2 | 2 | 0

2 | 1 | 0

0 | 1

$$(237)_8 = (159)_{10} = (10011111)_2$$

## Binary → Octal → Binary

| DEC | BIN | OCT |
|-----|-----|-----|
| 0   | 000 | 0   |
| 1   | 001 | 1   |
| 2   | 010 | 2   |
| 3   | 011 | 3   |
| 4   | 100 | 4   |
| 5   | 101 | 5   |
| 6   | 110 | 6   |
| 7   | 111 | 7   |

Example 2:

Convert  $(103)_8$  to Decimal

Solution:

$$\begin{aligned} (103)_8 &= (001\ 000\ 011)_2 \\ &= (1000011)_2 \end{aligned}$$

Validate the above answer mathematically

## Binary → Hexa → Binary

## Binary → Hexa → Binary

| Decimal | BIN  | HEX |
|---------|------|-----|
| 0       | 0000 | 0   |
| 1       | 0001 | 1   |
| 2       | 0010 | 2   |
| 3       | 0011 | 3   |
| 4       | 0100 | 4   |
| 5       | 0101 | 5   |
| 6       | 0110 | 6   |
| 7       | 0111 | 7   |
| 8       | 1000 | 8   |
| 9       | 1001 | 9   |
| 10      | 1010 | A   |
| 11      | 1011 | B   |
| 12      | 1100 | C   |
| 13      | 1101 | D   |
| 14      | 1110 | E   |
| 15      | 1111 | F   |

- The mathematical method of converting from binary to Hexa and vice versa is to convert the number first to Decimal
- The shortcut method is to cluster the binary number with 4-bit binary strings from the decimal point & then look up at the table to get its equivalent.

## Binary → Hexa → Binary

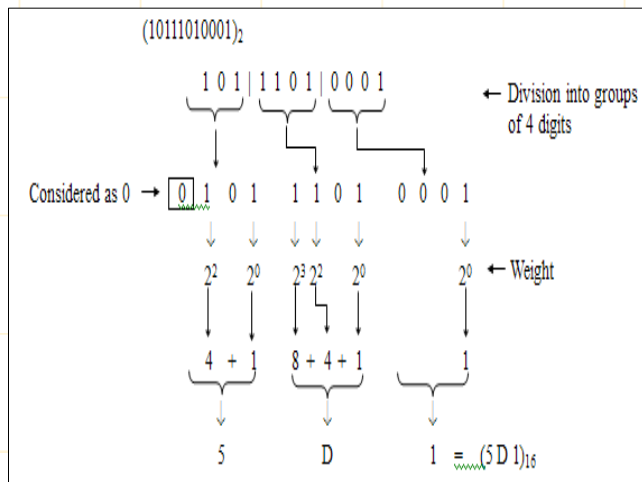
| Decimal | BIN  | HEX |
|---------|------|-----|
| 0       | 0000 | 0   |
| 1       | 0001 | 1   |
| 2       | 0010 | 2   |
| 3       | 0011 | 3   |
| 4       | 0100 | 4   |
| 5       | 0101 | 5   |
| 6       | 0110 | 6   |
| 7       | 0111 | 7   |
| 8       | 1000 | 8   |
| 9       | 1001 | 9   |
| 10      | 1010 | A   |
| 11      | 1011 | B   |
| 12      | 1100 | C   |
| 13      | 1101 | D   |
| 14      | 1110 | E   |
| 15      | 1111 | F   |

- In the event that there is a bit string with less than 4 digits, the necessary number of 0's is added and the string is considered as a 4-bit string.

## Binary → Hexa → Binary

| DEC | BIN  | HEX |
|-----|------|-----|
| 0   | 0000 | 0   |
| 1   | 0001 | 1   |
| 2   | 0010 | 2   |
| 3   | 0011 | 3   |
| 4   | 0100 | 4   |
| 5   | 0101 | 5   |
| 6   | 0110 | 6   |
| 7   | 0111 | 7   |
| 8   | 1000 | 8   |
| 9   | 1001 | 9   |
| 10  | 1010 | A   |
| 11  | 1011 | B   |
| 12  | 1100 | C   |
| 13  | 1101 | D   |
| 14  | 1110 | E   |
| 15  | 1111 | F   |

Example 1:



## Binary → Hexa → Binary

Example 1: Convert 10111010001 to HEX

*Solution (Mathematical):*

1. Convert BIN to DEC

$$10111010001 = 1 \cdot 2^{10} + 1 \cdot 2^8 + 1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^4 + 1 \cdot 2^0$$

$$10111010001 = 1489$$

2. Convert 1489 to Hex

$$(1489)_{10} = (5D1)_{16}$$

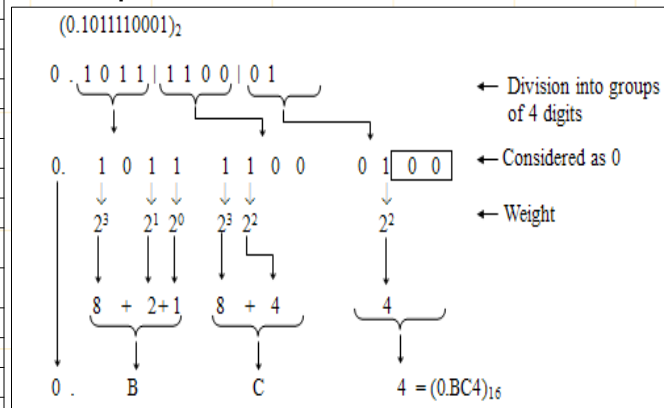
|    |      |     |
|----|------|-----|
| 16 | 1489 | Rem |
| 16 | 93   | 1   |
| 16 | 5    | 13  |
|    | 0    | 5   |



## Binary → Hexa → Binary

| DEC | BIN  | HEX |
|-----|------|-----|
| 0   | 0000 | 0   |
| 1   | 0001 | 1   |
| 2   | 0010 | 2   |
| 3   | 0011 | 3   |
| 4   | 0100 | 4   |
| 5   | 0101 | 5   |
| 6   | 0110 | 6   |
| 7   | 0111 | 7   |
| 8   | 1000 | 8   |
| 9   | 1001 | 9   |
| 10  | 1010 | A   |
| 11  | 1011 | B   |
| 12  | 1100 | C   |
| 13  | 1101 | D   |
| 14  | 1110 | E   |
| 15  | 1111 | F   |

Example 2:



Validate the above answer mathematically

## Binary → Hexa → Binary

| DEC | BIN  | HEX |
|-----|------|-----|
| 0   | 0000 | 0   |
| 1   | 0001 | 1   |
| 2   | 0010 | 2   |
| 3   | 0011 | 3   |
| 4   | 0100 | 4   |
| 5   | 0101 | 5   |
| 6   | 0110 | 6   |
| 7   | 0111 | 7   |
| 8   | 1000 | 8   |
| 9   | 1001 | 9   |
| 10  | 1010 | A   |
| 11  | 1011 | B   |
| 12  | 1100 | C   |
| 13  | 1101 | D   |
| 14  | 1110 | E   |
| 15  | 1111 | F   |

*Example 3:*

Convert  $(4A1)_{16}$  to Binary

*Solution:*

$4 \rightarrow 0100$

$A \rightarrow 1010$

$1 \rightarrow 0001$

*Therefore:*

$(4A1)_{16} = (0100\ 1010\ 0001)_2$

$= (100\ 1010\ 0001)_2$

## Binary → Hexa → Binary

*Example 3:* Convert  $(4A1)_{16}$  to Binary

*Solution (Mathematical):*

1. Convert HEX to DEC

$$(4A1)_{16} = 4 \cdot 16^2 + 10 \cdot 16^1 + 1 \cdot 16^0 = 1185$$

2. Convert 1185 to BIN

## Binary → Hexa → Binary

Example 3: Convert  $(4A1)_{16}$  to Binary

|   |      |     |
|---|------|-----|
| 2 | 1185 | Rem |
| 2 | 592  | 1   |
| 2 | 296  | 0   |
| 2 | 148  | 0   |
| 2 | 74   | 0   |
| 2 | 37   | 0   |
| 2 | 18   | 1   |
| 2 | 9    | 0   |

|   |   |   |
|---|---|---|
| 2 | 4 | 1 |
| 2 | 2 | 0 |
| 2 | 1 | 0 |
|   | 0 | 1 |

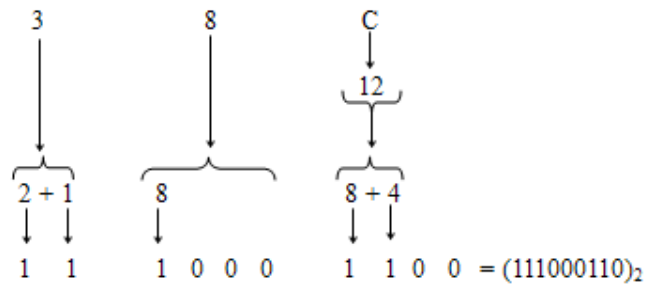
$$1185 = (10010100001)_2$$

## Binary → Hexa → Binary

Example 4: Convert  $(38C)_{16}$  to Binary

| DEC | BIN  | HEX |
|-----|------|-----|
| 0   | 0000 | 0   |
| 1   | 0001 | 1   |
| 2   | 0010 | 2   |
| 3   | 0011 | 3   |
| 4   | 0100 | 4   |
| 5   | 0101 | 5   |
| 6   | 0110 | 6   |
| 7   | 0111 | 7   |
| 8   | 1000 | 8   |
| 9   | 1001 | 9   |
| 10  | 1010 | A   |
| 11  | 1011 | B   |
| 12  | 1100 | C   |
| 13  | 1101 | D   |
| 14  | 1110 | E   |
| 15  | 1111 | F   |

$(38C)_{16}$

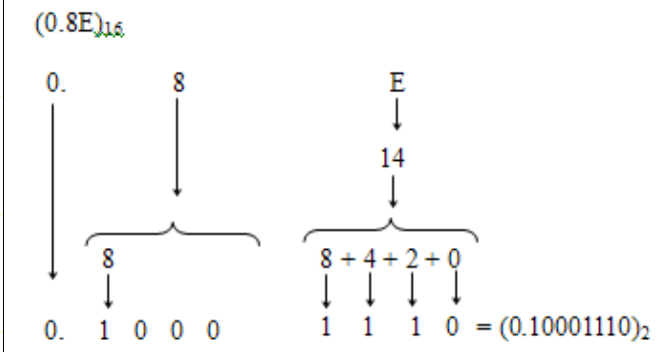


Validate the above answer mathematically

## Binary → Hexa → Binary

| DEC | BIN  | HEX |
|-----|------|-----|
| 0   | 0000 | 0   |
| 1   | 0001 | 1   |
| 2   | 0010 | 2   |
| 3   | 0011 | 3   |
| 4   | 0100 | 4   |
| 5   | 0101 | 5   |
| 6   | 0110 | 6   |
| 7   | 0111 | 7   |
| 8   | 1000 | 8   |
| 9   | 1001 | 9   |
| 10  | 1010 | A   |
| 11  | 1011 | B   |
| 12  | 1100 | C   |
| 13  | 1101 | D   |
| 14  | 1110 | E   |
| 15  | 1111 | F   |

Example 5:



Validate the above answer mathematically

Octal → Hexadecimal  
Hexadecimal → Octal

## Octal → Hexadecimal

Method 1:

1. Convert Octal → BIN  
 $(103)_8 = (001\ 000\ 011)_2$
2. Then group the bits by 4  
to produce the HEX value  
 $(\underline{0}\ \underline{0100}\ \underline{0011})_2 = (43)_{16}$

| DEC | BIN  | OCT | HEX |
|-----|------|-----|-----|
| 0   | 0000 | 0   | 0   |
| 1   | 0001 | 1   | 1   |
| 2   | 0010 | 2   | 2   |
| 3   | 0011 | 3   | 3   |
| 4   | 0100 | 4   | 4   |
| 5   | 0101 | 5   | 5   |
| 6   | 0110 | 6   | 6   |
| 7   | 0111 | 7   | 7   |
| 8   | 1000 | 10  | 8   |
| 9   | 1001 | 11  | 9   |
| 10  | 1010 | 12  | A   |
| 11  | 1011 | 13  | B   |
| 12  | 1100 | 14  | C   |
| 13  | 1101 | 15  | D   |
| 14  | 1110 | 16  | E   |
| 15  | 1111 | 17  | F   |

## Octal → Hexadecimal

Method 2 (Mathematical): Convert OCT → DEC → HEX

Example: Convert  $(103)_8$  to HEX

1. Convert OCT → DEC.  $(103)_8 = 1 \cdot 8^2 + 3 = 67$
2. Then convert DEC → OCT
3.  $(103)_8 = (67)_{10} = (43)_{16}$

|    |    |     |
|----|----|-----|
| 16 | 67 | Rem |
| 16 | 4  | 3   |
|    | 0  | 4   |



## Hexadecimal → Octal

Method 1:

1. Convert HEX – BIN  
 $(43)_{16} = (\underline{0100} \underline{0011})_2$
2. Then group the bits by 3 from the right to produce the octal value  
 $(\underline{1} \underline{000} \underline{011})_2 = (103)_8$

| DEC | BIN  | OCT | HEX |
|-----|------|-----|-----|
| 0   | 0000 | 0   | 0   |
| 1   | 0001 | 1   | 1   |
| 2   | 0010 | 2   | 2   |
| 3   | 0011 | 3   | 3   |
| 4   | 0100 | 4   | 4   |
| 5   | 0101 | 5   | 5   |
| 6   | 0110 | 6   | 6   |
| 7   | 0111 | 7   | 7   |
| 8   | 1000 | 10  | 8   |
| 9   | 1001 | 11  | 9   |
| 10  | 1010 | 12  | A   |
| 11  | 1011 | 13  | B   |
| 12  | 1100 | 14  | C   |
| 13  | 1101 | 15  | D   |
| 14  | 1110 | 16  | E   |
| 15  | 1111 | 17  | F   |

## Hexadecimal → Octal

Method 2 (Mathematical): Convert HEX → DEC → OCT

Example: Convert  $(43)_{16}$  to Octal

1. Convert HEX → DEC.  $(43)_{16} = 4 \cdot 16 + 3 = 67$
2. Then convert DEC → OCT
3.  $(43)_{16} = (67)_{10} = (103)_8$

|   |    |     |
|---|----|-----|
| 8 | 67 | Rem |
| 8 | 8  | 3   |
| 8 | 1  | 0   |
|   | 0  | 1   |



## References:

1. What is a computer? [www.webopedia.com](http://www.webopedia.com)
2. <http://perugini.cps.udayton.edu/teaching/courses/Spring2014/cps346/>
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5. Decimal Number System. <http://www.britannica.com/topic/decimal-number-system>

Thank you!